



CAMPUS SOS — AI EMERGENCY COORDINATION SYSTEM

PRESENTED BY TEAM MIND MATRIXX | CODE • INNOVATE • ELEVATE

EXECUTIVE SUMMARY: Campus SOS is an autonomous multi-agent emergency coordination system designed to eliminate critical response delays during campus crises. When a student reports an emergency in plain language, Campus SOS deploys a swarm of 5 specialized AI agents to analyze threats, map locations, dispatch security officers via WhatsApp, route campus ambulances, and notify medical staff and wardens.

1. The Human Problem Statement

Imagine a student at 11:30 PM in a university hostel:

'My roommate collapsed on the floor. We don't know whether to call security, call an ambulance, contact the warden, or call their parents.'

Core Weaknesses of Existing Systems:

- **Panic & Confusion:** Students must figure out the complex response chain manually during moments of extreme stress.
- **Alert Fatigue:** Traditional SOS apps send generic text alerts with zero coordination between responders.
- **Siloed Communication:** Security, medical staff, hostel wardens, and drivers operate on disconnected channels.

2. Vision & Key Differentiators

Traditional Campus Apps	Campus SOS (TEAM MIND MATRIXX)
'Our AI detects emergencies.'	'Our AI coordinates the humans & systems that respond to the emergency.'
Sends static push notification / SMS	Dispatches 5 autonomous specialized AI agents in parallel
Requires manual phone calls	Automated real WhatsApp message delivery (+916385710907 / +917639277606)
No visual campus tracking	Interactive satellite map with animated response routes & node selection

Uncontrolled automated actions	Human-in-the-Loop Supervisor approval gate with 30s countdown ring
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3. Multi-Agent Solution Architecture

Agent	Responsibility	Action Output
■ Emergency Agent	Parses raw natural language text into structured intent & severity.	Classifies type (Medical, Fire, Trauma, Security), severity (P1/P2), & first-aid steps.
■ Location Agent	Resolves campus node coordinates & calculates shortest paths.	Highlights target building (e.g. Block C) on map & draws animated routes.
■ Security Agent	Dispatches campus security officers & alerts patrol team.	Delivers real WhatsApp dispatch briefing to +91 7639277606.
■ Transport Agent	Scans vehicle fleet (Ambulance, Golf Carts) & calculates ETA.	Dispatches campus ambulance & triggers external 108 EMS escalation.
■ Communication Agent	Executes multi-stakeholder notification chain.	Sends real WhatsApp messages to Doctor (+91 6385710907), Wardens, & Custom Contacts.

4. Supervisor Gate & Tech Stack

- **Human-in-the-Loop Supervisor Gate:** Sensitive actions (external 108 ambulance call, parent notifications) require human authorization via an interactive 30-second countdown ring modal. For critical emergencies, safety auto-escalation approves action if unhandled.
- **Real WhatsApp Web API Integration:** Built with whatsapp-web.js, automatically delivering formatted markdown messages to real phone numbers.
- **Tech Stack:** Node.js, Express, WebSockets (ws), Google Gemini AI API, HTML5/CSS3 Cyberpunk Glassmorphism UI, SVG Satellite Mapping.